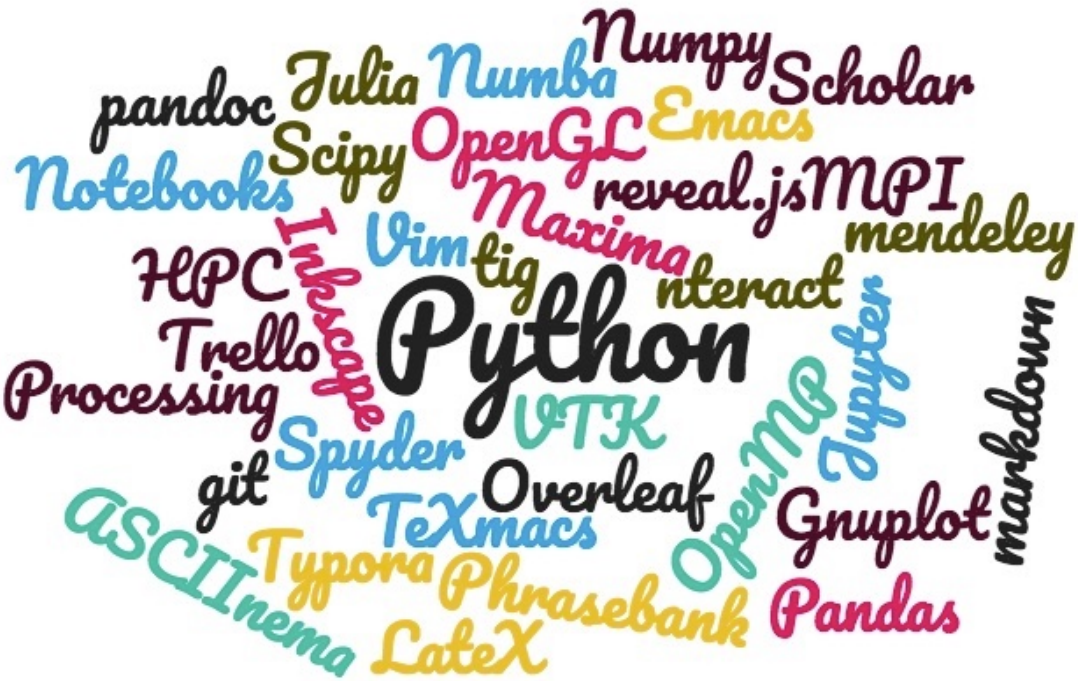
The efficient toolbox of the Computational Scientist



Dr. Gábor Závodszky - *Computational Science Lab*

https://github.com/gzavo/CS_Assignment

> "Never do a live demo" -- Every presenter ever

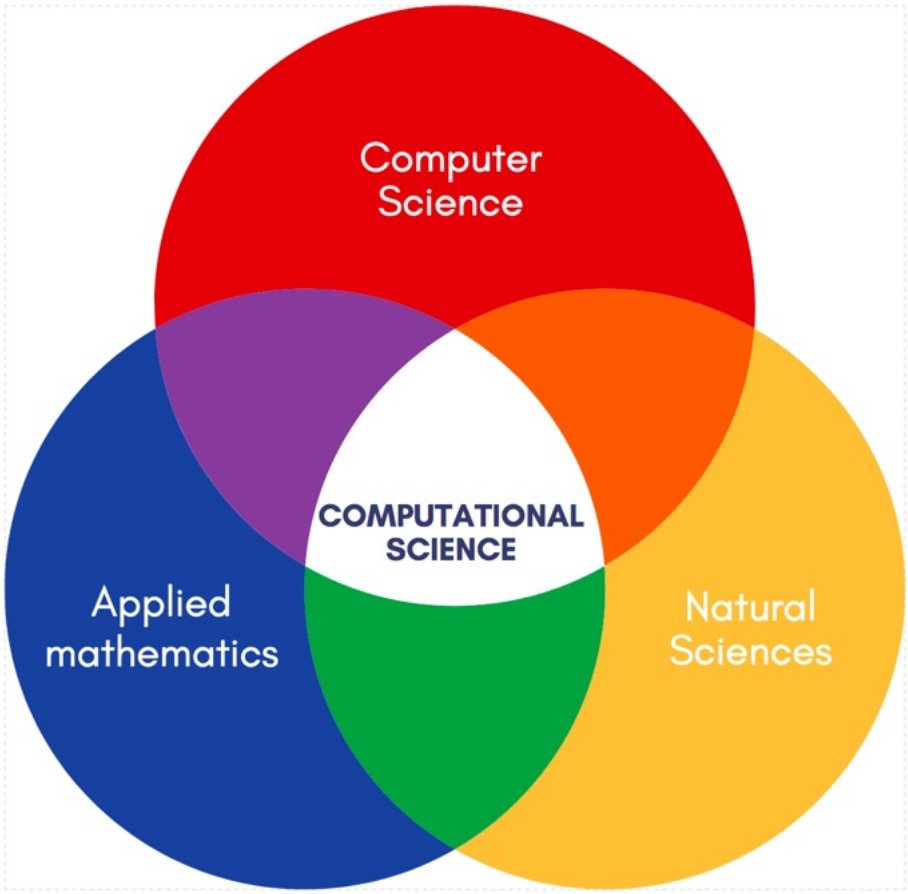
Structure of the lecture

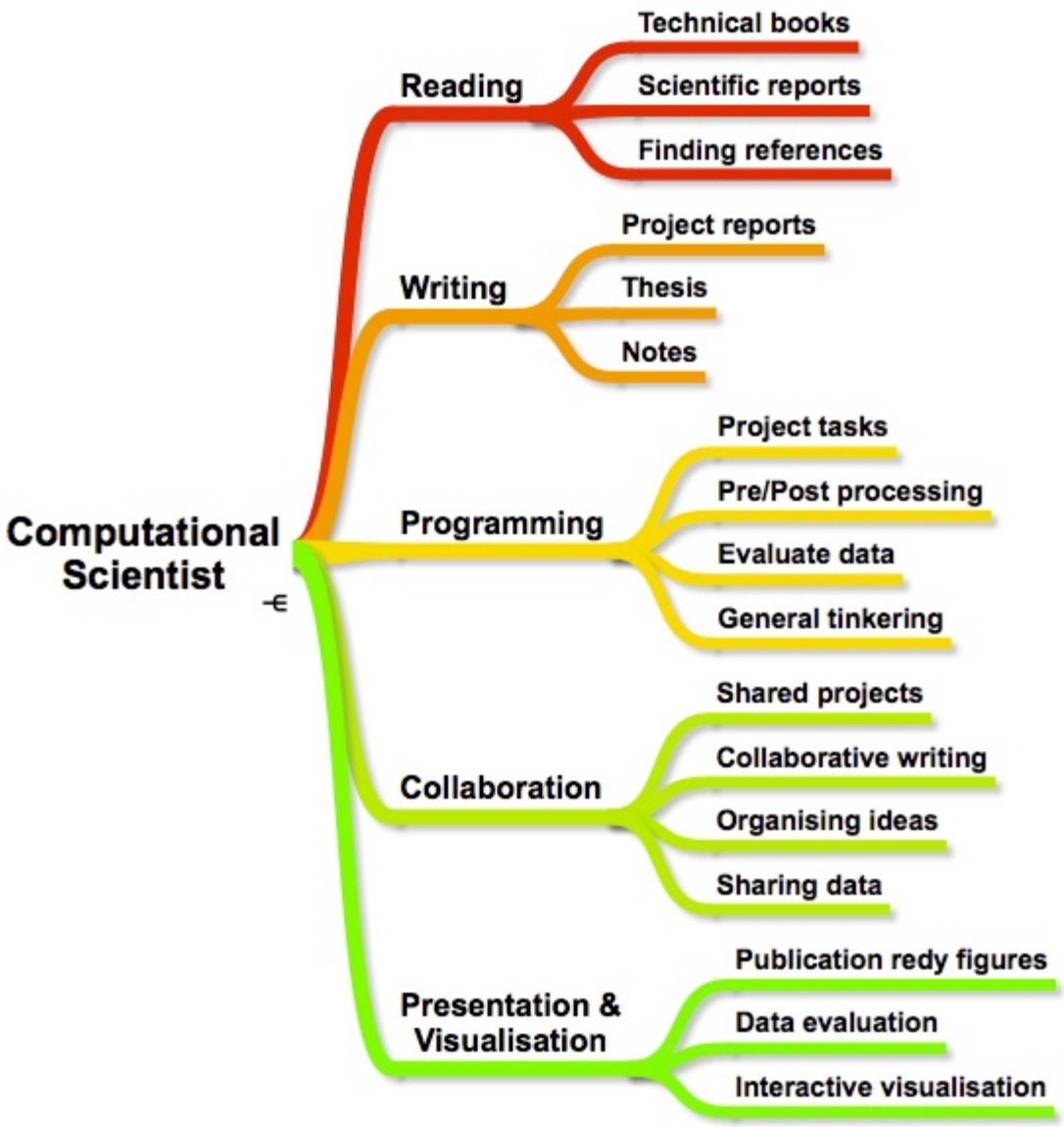
- Domains of "The Computational Scientist"
- Scientific writing and reading
- Programming and visualization basics
- Software development in a collaborative environment
- Homework assingment

- Can be an information overload!
- These slides contain the very basics.
- Due to the structure of this MSc programme, you might not need *everything*.

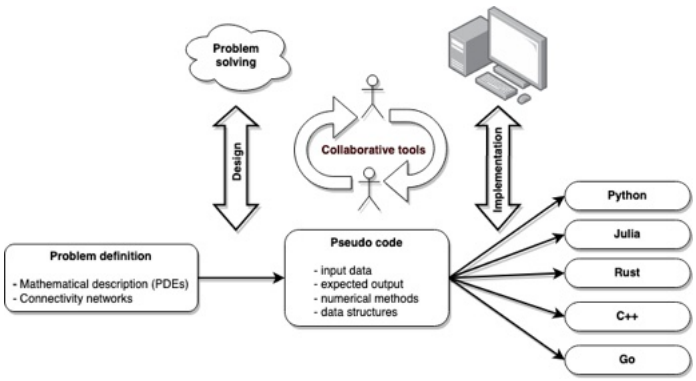
- The most important things are marked: **(!)**.
- Some things are there to hear at least once.
- You can use the pdf version as a list of useful tools.

Who is the Computational Scientist? - Domains and fundamental skills





General workflow of the Computational Scientist



CLS Activities - Reading

- UvA Library (!) (http://uba.uva.nl/en)
- You can also use it to get access to non-open access journal papers.

- Google Scholar (!) - https://scholar.google.com
  - Search for papers
  - export references
  - look at researcher profiles
  - E.g.: "Carbon monoxide, boldly goes where..."
- Web of Science, Scopus, ...

- Managing references: Mendeley, **Zotero** (!), Papers, etc.
  - Sync. paper database
  - Annotate pdfs, add notes and sync them as well
- Most students use Zotero (I use it as well).
- Mendeley gives more cloud space, but its less flexible.

CLS Activities - Writing

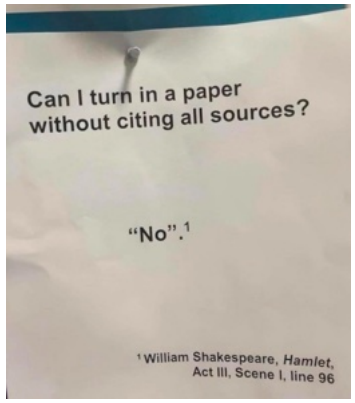
- LaTeX
  - Overleaf (!) - https://www.overleaf.com/
  - TexMaker
  - TeXmacs
  - latex2png - http://latex2png.com/
- Typist (modern LaTeX alternative written in Rust) https://typst.app/

-> Why is LaTeX useful?

- Not a WYSIWYG solution
- But often WYGIWYN
- Abundance of templates
- Literally a programming language for word processing

In [ ]:

Remember...



Collaborative writing

- Markdown (!)
  - Use Pandoc (!) to turn it to .doc, .pdf, .html, you name it (https://pandoc.org/)
  - Zettlr (!) (https://www.zettlr.com/)
  - Typora (https://typora.io/)
  - Mark Text (https://github.com/marktext/marktext)
  - Basically everywhere (websites, forums, editors...really, everywhere)
  - Other supersets / alternatives (ASCIIDOC, ...)

Markdown

Write equations: $e^{i\pi} + 1 = 0$

Embed code:

```
def f(x):
    """docstring of this very useful function"""
    return x**2
```

Add tables:

This	is
a	table

Create lists:

- list item 1
- list item 2
- list item 3

Markdown syntax

Markdown is a simple way to format text that looks great on any device. It doesn't do anything fancy like change the font size, color, or type — just the essentials, using keyboard symbols you already know.

TRY OUR 10 MINUTE MARKDOWN TUTORIAL

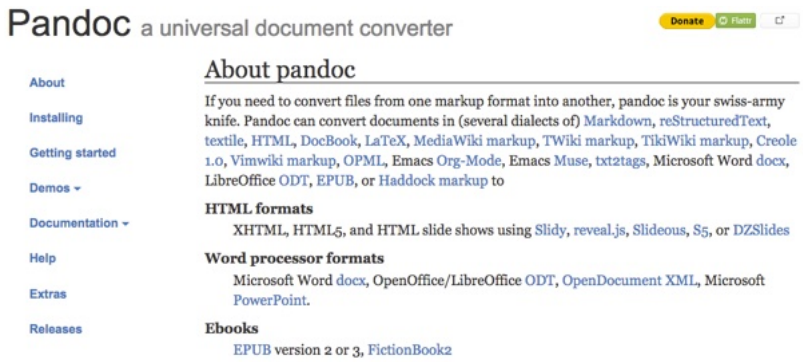
Type	Or	... to Get
`*Italic*`	`_Italic_`	*Italic*
`**Bold**`	`__Bold__`	**Bold**
`# Heading 1`	`Heading 1` `=========`	Heading 1
`## Heading 2`	`Heading 2` `---------`	Heading 2

`[Link](http://a.com)`	`[Link][1]`
	⋮
	`[1]: http://b.org`
		Link

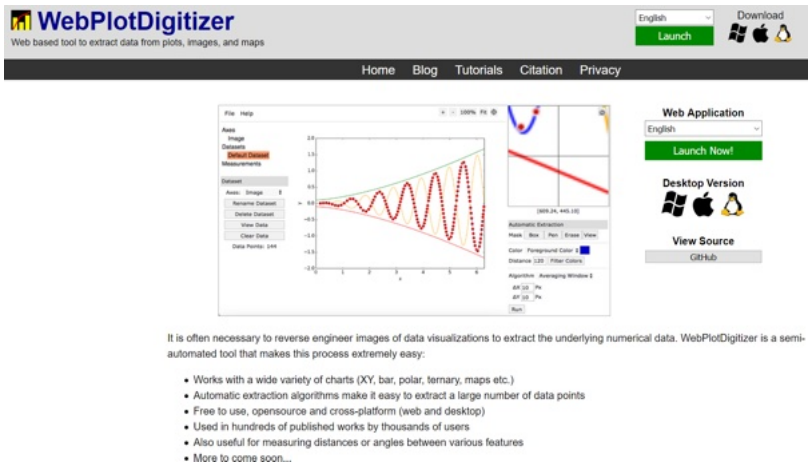
`![Image](http://url/a.png)`	`![Image][1]`
	⋮
	`[1]: http://url/b.jpg`
		M↓

`> Blockquote`		Blockquote
`* List` `* List` `* List`	`- List` `- List` `- List`	• List • List • List
`1. One` `2. Two`	`1) One` `2) Two`	1. One 2. Two

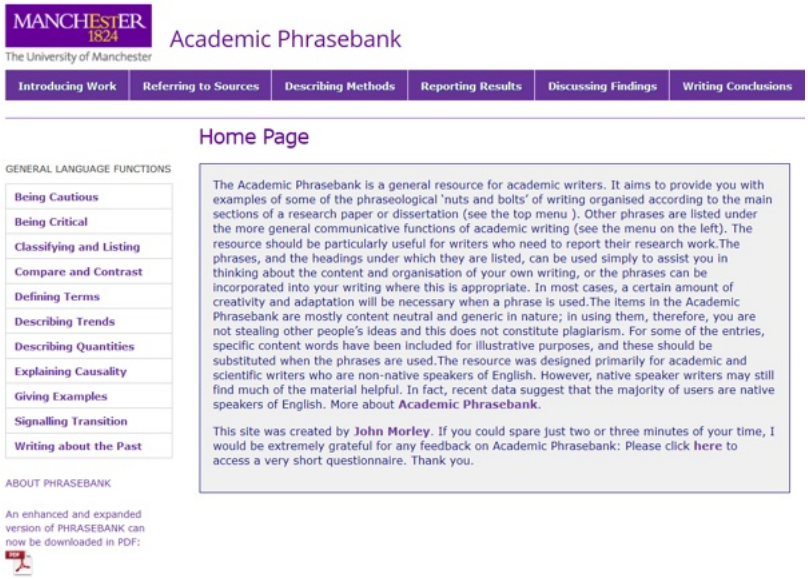
CLS Activities - Writing - cont.



https://automeris.io/WebPlotDigitizer/



http://www.phrasebank.manchester.ac.uk/



CLS Activities - Programming

- programming languages
- editors

- examples

Programming languages

- Python (!)
- Julia (!) (C + Python + Lisp structures)
- C (!) (performance, hardware access)
- C++ (!) (If you need performance + OO)
- Rust (!) (performance and safety)
- Kotlin (primarily mobile development)
- Nim (Python, but C)
- R (statistics)
- Scala (data science)
- Go (Google's try)
- Javascript (too popular to leave out)
- Racket (for the gourmet)
- +1 Lobster lang. (http://strlen.com/lobster/)

> Application domains can overlap, choose 2-3 and learn them well!

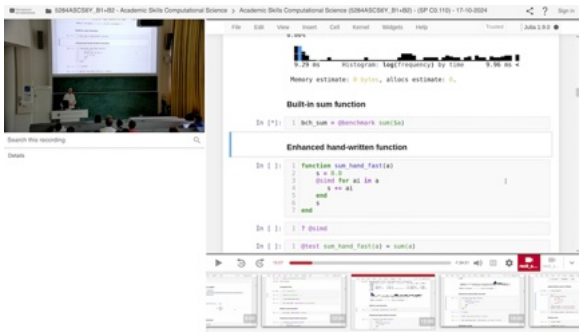
> For instance, C/C++ and Python is a quite versatile combination.

> Computer Language Benchmark Game (https://benchmarksgame-team.pages.debian.net/benchmarksgame/index.html)

> Rosetta Code (http://www.rosettacode.org)

Julia programming language

https://hva-uva.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=00d4221b-42c8-4d3b-978b-b1e900b401d3



In [ ]:

General editors

- **Visual Studio Code** (!!)
- Zed (Editor, AI support) - https://zed.dev/
- Gram (Zed without AI)
- **Jupyter** / **JupyterLab** (!)
- Cursor, Roo, ... zoo of AI assisted IDEs
- Vi/Vim/Neo Vim
- Emacs (Spacemacs)
- Sublime Text
- Note++
- ... wide palett of other editors ...

Specific editors / IDEs

- Spyder (!) - https://www.spyder-ide.org/
- **Pluto** (for Julia) (!) - https://plutojl.org/
- PyCharm
- Juno (VSCode, Julia) - https://www.julia-vscode.org/
- Lazarus (UI for desktop) - https://www.lazarus-ide.org/

In practice

Why is Jupyter important?

- HTML5 platform (kernel as backend, can run remotely on a different machine)
- Compatible with several languages (C++, Python, Julia, Haskell, ...)
- Important step in hosting and sharing codes (reproducability)
- Towards reproducable science! (also see Jupyter Lab)

> Careful: non-linear state, see next example! (Check out Pluto for linear state solution)

- Additional benefits of the separate backend: parallel execution, cluster management, remote execution

In [1]:
```
b=2
print(b)
```

2

In [2]:
```
print(b)
b=3
```

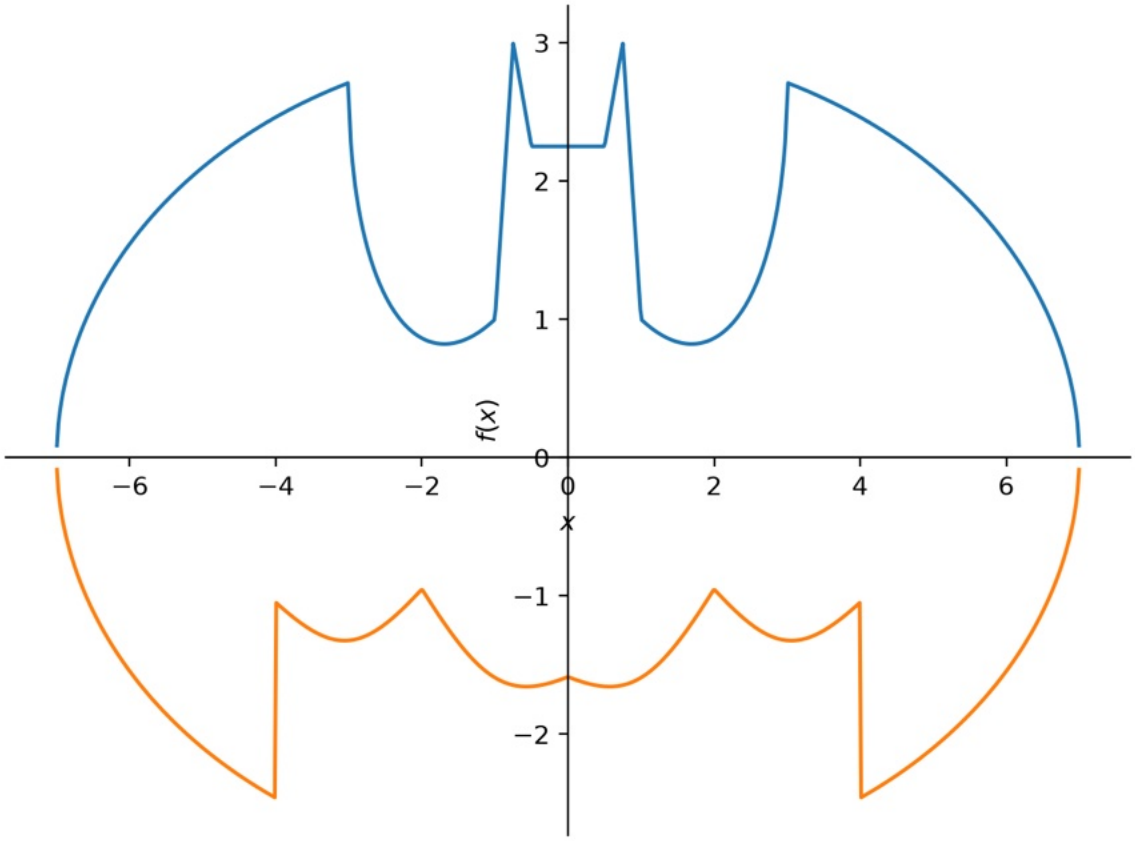
2

In [3]:
```
%matplotlib notebook
import sympy as sp
sp.init_printing(use_latex='mathjax')
x,y,z = sp.symbols('x,y,z')
f = sp.sin(x*y)+sp.cos(y*z)
sp.integrate(f,x)
```

Out[3]:

$$x\cos(yz) + \begin{cases} -\frac{\cos(xy)}{y} & \text{for } y \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

In [4]:
```
import sympy as sm
from sympy.plotting import plot
x = sm.symbols('x', real=True)
h_ = sm.symbols('h_')

w = 3 * sm.sqrt(1 - (x / 7)**2)
l = ((x + 3) / 2 - sm.S(3) / 7 * sm.sqrt(10) * sm.sqrt(4 - (x + 1)**2) +
    sm.S(6) / 7 * sm.sqrt(10))
r = ((3 - x) / 2 - sm.S(3) / 7 * sm.sqrt(10) * sm.sqrt(4 - (x - 1)**2) +
    sm.S(6) / 7 * sm.sqrt(10))
h = (3*(sm.Abs(x - sm.S.Half) + sm.Abs(x + sm.S.Half) + 6) -
              11*(sm.Abs(x - sm.S(3)/4) + sm.Abs(x + sm.S(3)/4)))/2
f = ((h - l) * sm.Heaviside(x + 1, 0) +
    (r - h) * sm.Heaviside(x - 1, 0) +
    (l - w) * sm.Heaviside(x + 3, 0) +
    (w - r) * sm.Heaviside(x - 3, 0) +
    w)
g = (sm.S(1) / 2 * (sm.Abs(x / 2) + sm.sqrt(1 + (sm.Abs(sm.Abs(x) - 2) -
    1)**2) - sm.S(1) / 112 * (3 * sm.sqrt(33) - 7) * x**2 + 3 *
    sm.sqrt(1 - (sm.S(1)/ 7 * x)**2) - 3) * ((x + 4) / sm.Abs(x + 4) -
    (x - 4) / sm.Abs(x - 4)) - 3 * sm.sqrt(1 - (x / 7)**2))
sm.plot(f, g)
```



Out[4]: <sympy.plotting.backends.matplotlibbackend.matplotlib.MatplotlibBackend at 0x13fc49b50>

SymPy Cheatsheet (http://sympy.org)

Basics

Sympy help:	`help(function)`
Declare symbol:	`x = Symbol('x')`
Substitution:	`expr.subs(old, new)`
Numerical evaluation:	`expr.evalf()`
Expanding:	`expr.expand()`
Common denominator:	`ratsimp(expr)`
Simplify expression:	`simplify(expr)`

Constants

π:	`pi`
e:	`E`
∞:	`oo`
i:	`I`

Numbers types

Integers ($\mathbb{Z}$):	`Integer(x)`
Rationals ($\mathbb{Q}$):	`Rational(p, q)`
Reals ($\mathbb{R}$):	`Float(x)`

Basic funtions

Trigonometric:	`sin cos tan cot`
Cyclometric:	`asin acos atan acot`
Hyperbolic:	`sinh cosh tanh coth`
Area hyperbolic:	`asinh acosh atanh acoth`
Exponential:	`exp(x)`
Square root:	`sqrt(x)`
Logarithm ($\log_b a$):	`log(a, b)`
Natural logarithm:	`log(a)`
Gamma ($\Gamma(x)$):	`gamma(x)`
Absolute value:	`abs(x)`

Calculus

$\lim_{x\to a} f(x)$:	`limit(f, x, a)`
$\lim_{x\to a_-} f(x)$:	`limit(f, x, a, dir='-')`
$\lim_{x\to a_+} f(x)$:	`limit(f, x, a, dir='+')`
$\frac{d}{dx} f(x)$:	`diff(f, x)`
$\frac{\partial}{\partial x} f(x, y)$:	`diff(f, x)`
$\int f(x)\,dx$:	`integrate(f, x)`
$\int_a^b f(x)\,dx$:	`integrate(f, (x, a, b))`
Taylor series (at a, deg n)	`f.series(x, a, n)`

Equations

Equation $f(x) = 0$:	`solve(f, x)`
System of equations:	`solve([f, g], [x, y])`
Differential equation:	`dsolve(equation, f(x))`

Geometry

Points:	`a = Point(xcoord, ycoord)`
Lines:	`l = Line(pointA, pointB)`
Circles:	`c = Circle(center, radius)`
Triangles:	`t = Triangle(a, b, c)`
Area:	`object.area`
Intersection:	`intersection(a, b)`
Checking tangency:	`c.is_tangent(l)`

Plotting

Plot:	`Plot(f, [a, b])`
Zoom: +/−:	R/F or PgUp/PgDn or Numpad +/-
Rotate X,Y axis:	Arrow Keys or WASD
Rotate Z axis:	Q and E or Numpad 7 and 9
View XY:	F1
View XZ:	F2
View YZ:	F3
View Perspective:	F4
Axes Visibility:	F5
Axes Colors:	F6
Screenshot:	F8
Exit plot:	ESC

Discrete math

Factorial ($n!$):	`factorial(n)`
Binomial coefficient $\binom{n}{k}$:	`binomial(n, k)`
Sum ($\sum_{n=a}^{b} expr$):	`summation(expr, (n, a, b))`
Product ($\prod_{n=a}^{b} expr$):	`product(expr, (n, a, b))`

Linear algebra

Matrix definition:	`m = Matrix([[a, b], [c, d]])`
Determinant:	`m.det()`
Inverse:	`m.inv()`
Identity matrix $n \times n$:	`eye(n)`
Zero matrix $n \times n$:	`zeros(n)`
Ones matrix $n \times n$:	`ones(n)`

Printing

LaTeX print:	`print latex()`
Python print:	`print python()`
Pretty print:	`pprint()`

Examples

Find 100 digits of π^e:
`(pi**E).n(100)`

Expand $(x+y)^2(x-y)(x^2+y)$:
`((x + y)**2 * (x - y) * (x**2 + y)).expand()`

Simplify $\frac{1}{x} + \frac{x \sin x - 1}{x^2 - 1}$:
`simplify((1/x) + (x * sin(x) - 1)/(x**2 - 1))`

Check if line passing through points $(0,1)$ and $(1,1)$ is tangent to circle with center at $(5,5)$ and radius 3:
`Circle(Point(5,5), 3).is_tangent(Line(Point(0,1), Point(1,1)))`

Find roots of $x^4 - 4x^3 + 2x^2 - x = 0$:
`solve(x**4 - 4*x**3 + 2*x**2 - x, x)`

Solve the equations system: $x + y = 4$, $xy = 3$:
`solve([x + y - 4, x*y - 3], [x, y])`

Calculate limit of the sequence $\sqrt[n]{n}$:
`limit(n**(1/n), n, oo)`

Calculate left-sided limit of the function $\frac{|x|}{x}$ in 0:
`limit(abs(x)/x, x, 0, dir='-')`

Calculate the sum $\sum_{n=0}^{100} n^2$:
`summation(n**2, (n, 0, 100))`

Calculate the sum $\sum_{n=0}^{\infty} \frac{1}{n^2}$:
`summation(1/n**2, (n, 0, oo))`

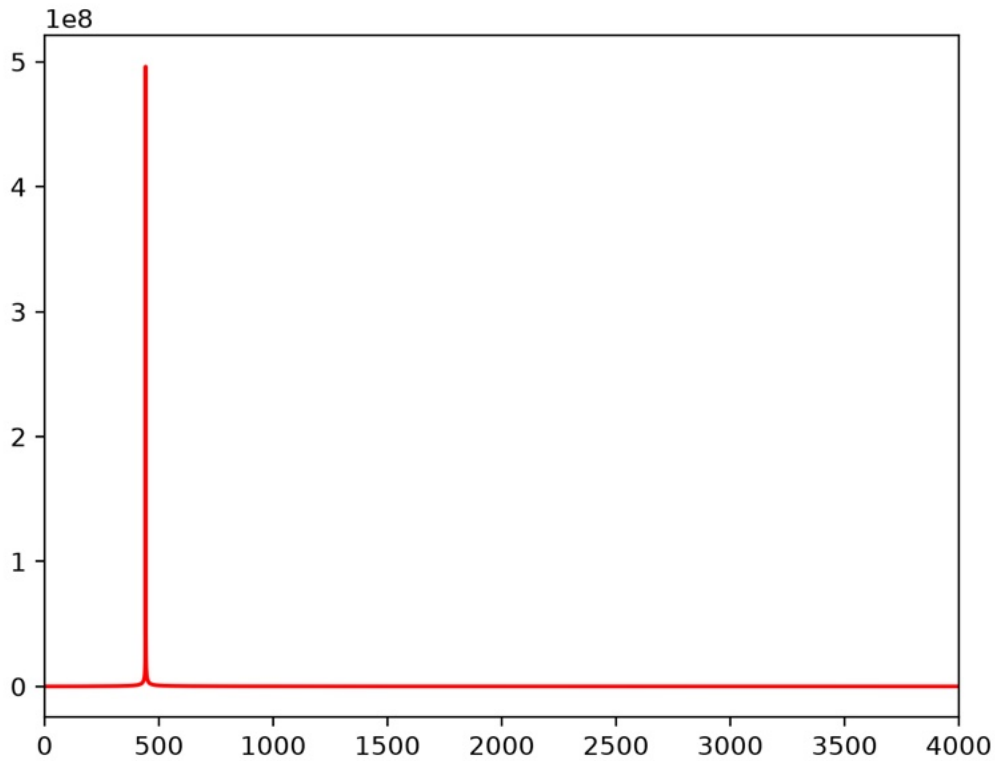
Calculate the integral $\int \cos^3 x\,dx$:
`integrate(cos(x)**3, x)`

Calculate the integral $\int_1^{\infty} \frac{dx}{x^2}$:
`integrate(1/x**2, (x, 1, oo))`

Find 10 terms of series expansion of $\frac{1}{1-2x}$ at 0:
`(1/(1 - 2*x)).series(x, 0, 10)`

Solve the differential equation $f''(x) + 9f(x) = 1$:
`dsolve(f(x).diff(x, x) + 9*f(x) - 1, f(x))`

In [5]:
```
import matplotlib.pyplot as plt
from scipy.fftpack import fft; from scipy.io import wavfile
fs, data = wavfile.read('sound/sine440.wav') # load the data, 16 bit, 44.1 k
c = fft(data.T) # calculate fourier transform (complex numbers list)
d = len(c)//2  # you only need half of the fft list (real signal symmetry)
plt.figure(); plt.plot(abs(c[:(d-1)]),'r'); plt.xlim((0,4000))
```

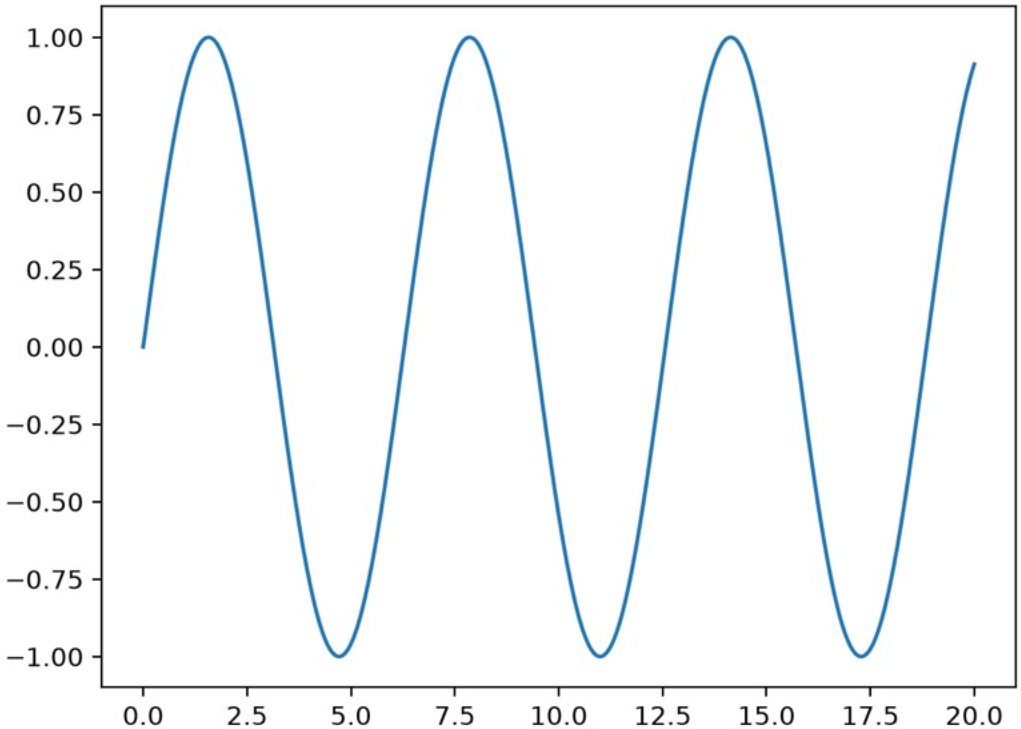


Out[5]: (0.0, 4000.0)

In [6]:
```
import IPython
IPython.display.Audio('sound/sine440.wav')
# IPython.display.Audio('sound/sample.wav')
```

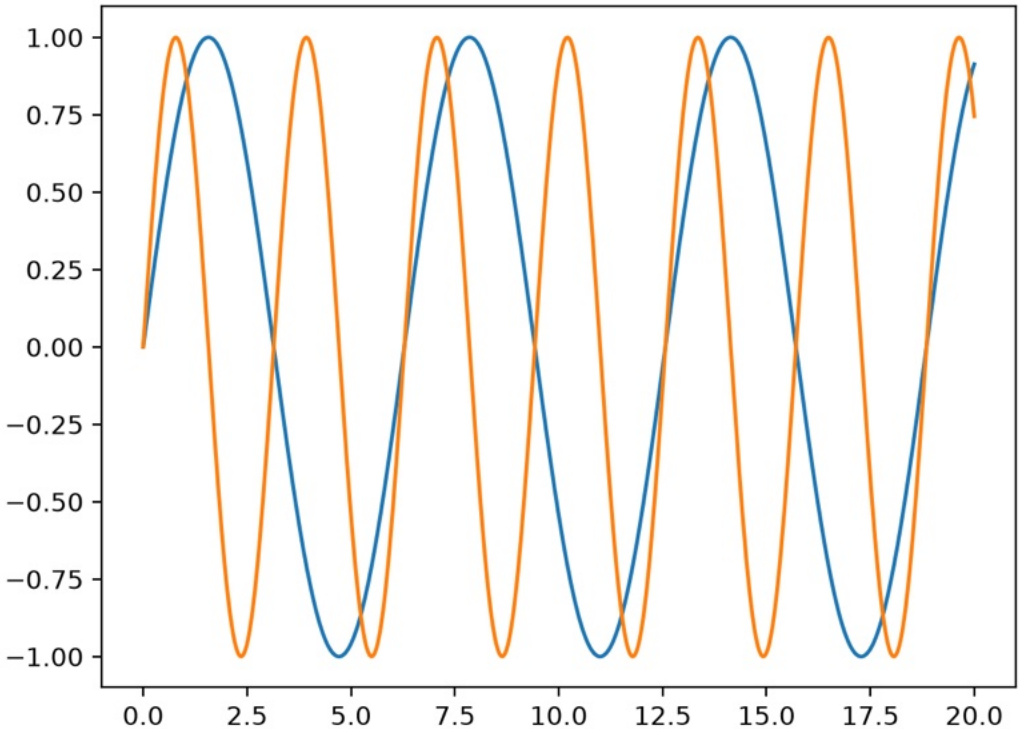
Out[6]: 0:00 / 0:01

In [7]:
```
import matplotlib.pyplot as plt
import numpy as np
x = np.linspace(0,20,500)
plt.figure(); plt.plot(x, np.sin(x))
```



Out[7]: [<matplotlib.lines.Line2D at 0x15a380cb0>]

In [8]:
```
import matplotlib.pyplot as plt
from ipywidgets import interact, IntSlider
from IPython.display import display,clear_output
plt.figure()

def f(freq):
    x = np.linspace(0,20,500)
    plt.plot(x, np.sin(x*freq))
    plt.show()
    #display(plt.figure)

interact(f, freq=IntSlider(min=1,max=5,step=1,value=1));
```



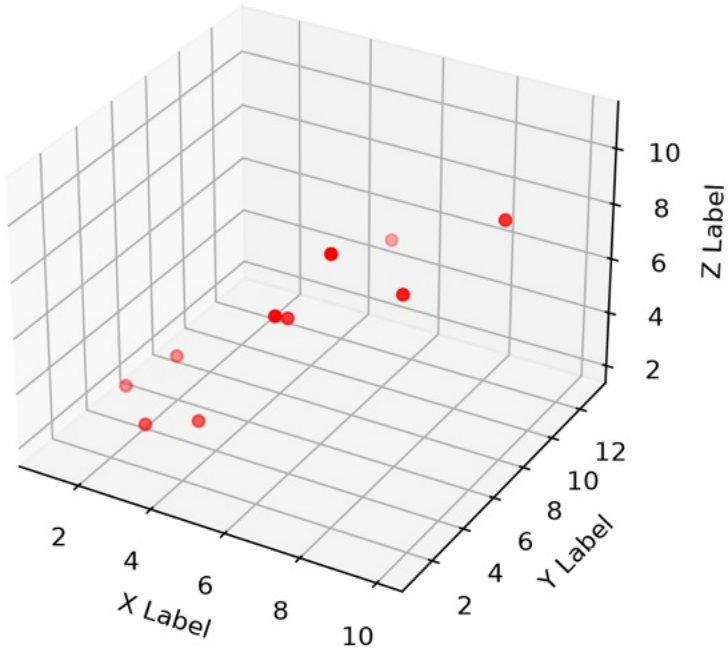

```
interactive(children=(IntSlider(value=1, description='freq', max=5, min=1),
Output()), _dom_classes=('widget-i…
```

In [9]:
```
from IPython.display import display, Javascript
import ipywidgets as widgets

L = widgets.Label("Hello World")
display(L)
```

```
Label(value='Hello World')
```

In [10]:
```
L.value = "Howdy World"
```

In [11]:
```
from mpl_toolkits.mplot3d import Axes3D; import matplotlib.pyplot as plt;

fig = plt.figure(); ax = fig.add_subplot(111, projection='3d')
x =[1,2,3,4,5,6,7,8,9,10]; y =[5,6,2,3,13,4,1,2,4,8]; z =[2,3,3,3,5,7,9,11,9
ax.scatter(x, y, z, c='r', marker='o'); ax.set_xlabel('X Label'); ax.set_yla
```



Out[11]: Text(0.5, 0, 'Z Label')

In [ ]:
```
import ipyvolume as ipv
import numpy as np
import ipyvolume.datasets
stream = ipyvolume.datasets.animated_stream.fetch()
fig = ipv.figure()
# instead of doing x=stream.data[0], y=stream.data[1], ... vz=stream.data[5]
# limit to 50 timesteps to avoid having a huge notebook
q = ipv.quiver(*stream.data[:,0:50,:200], color="red", size=7)
ipv.style.use("dark") # looks better
ipv.animation_control(q, interval=200)
ipv.show()
```

In [12]:
```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
df = sns.load_dataset("anscombe") #Anscombe's quartet, there are others (Tit
df.head()
```

Out[12]:

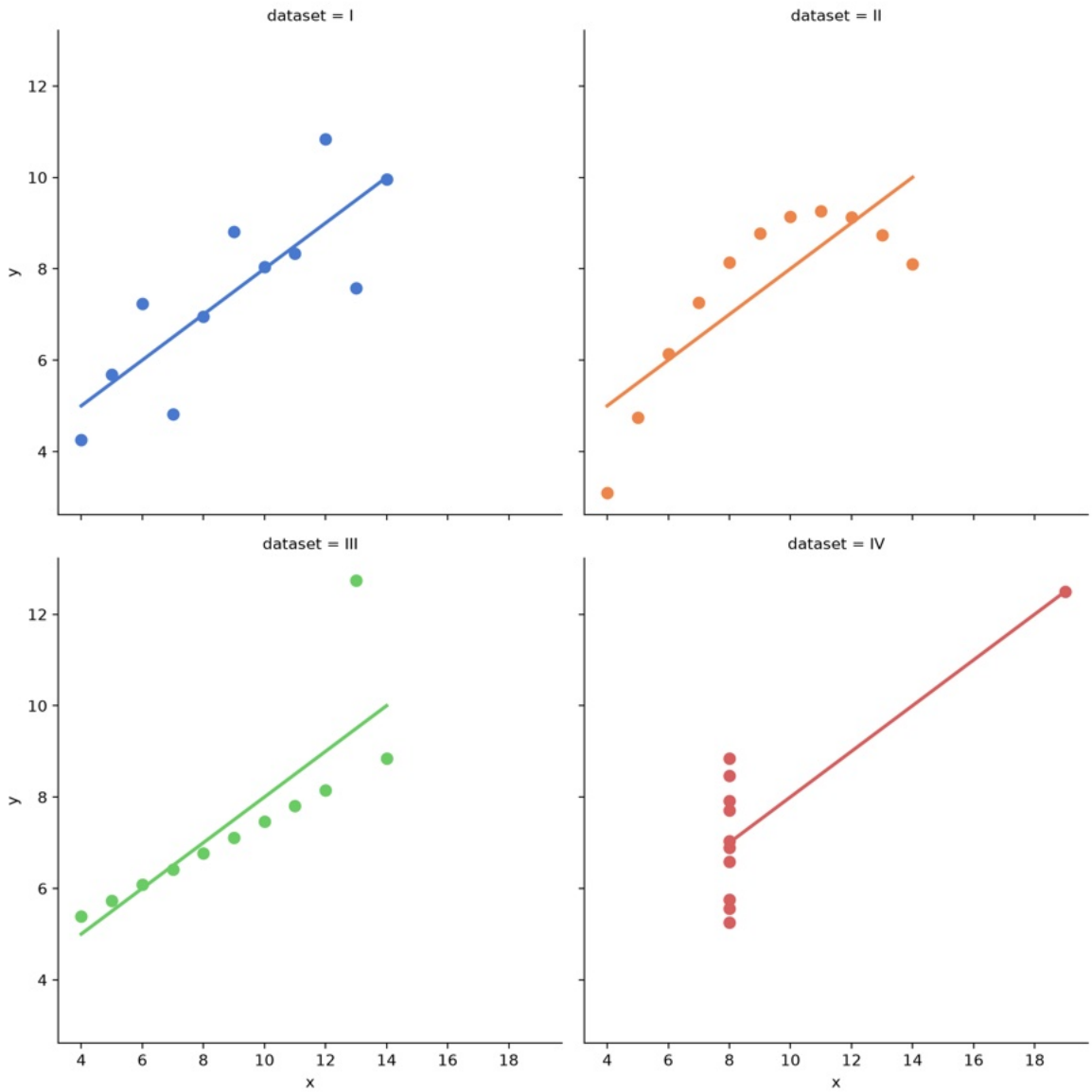
	dataset	x	y
0	I	10.0	8.04
1	I	8.0	6.95
2	I	13.0	7.58
3	I	9.0	8.81
4	I	11.0	8.33

In [13]:
```
df.groupby(df.dataset).describe()
```

Out[13]:

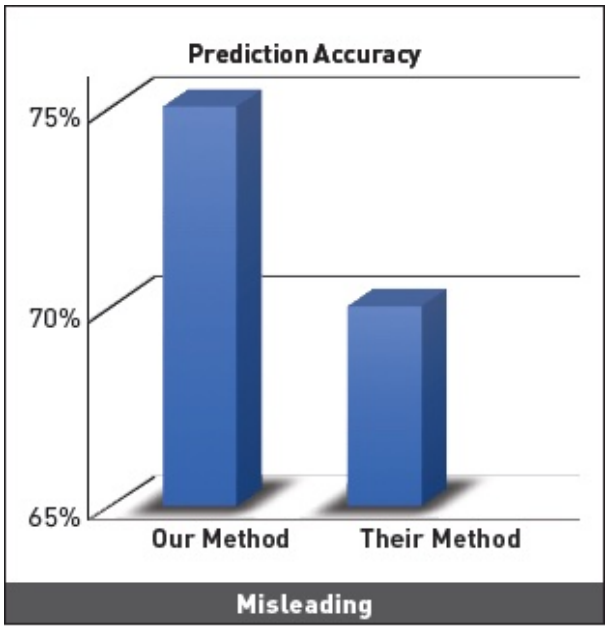
								x			
	count	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**	**count**	**mean**	
dataset											
I	11.0	9.0	3.316625	4.0	6.5	9.0	11.5	14.0	11.0	7.500909	2.031
II	11.0	9.0	3.316625	4.0	6.5	9.0	11.5	14.0	11.0	7.500909	2.031
III	11.0	9.0	3.316625	4.0	6.5	9.0	11.5	14.0	11.0	7.500000	2.030
IV	11.0	9.0	3.316625	8.0	8.0	8.0	8.0	19.0	11.0	7.500909	2.030

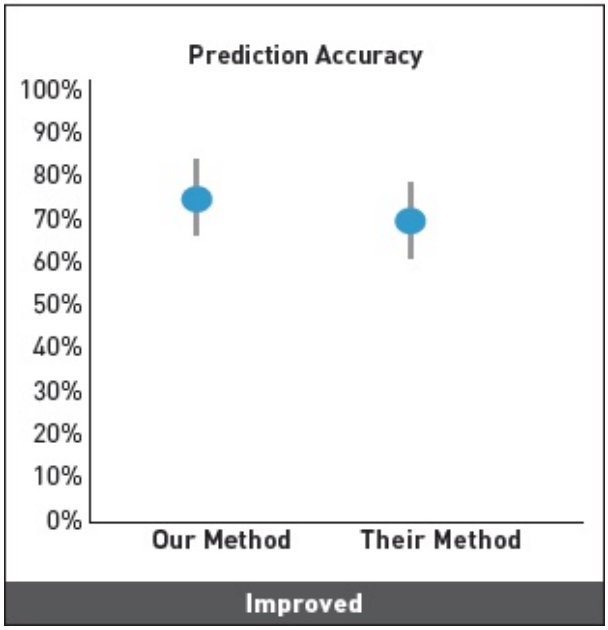
In [14]:
```
sns.lmplot(x="x", y="y", col="dataset", hue="dataset", data=df, col_wrap=2,
           scatter_kws={"s": 50, "alpha": 1}); plt.show()
```

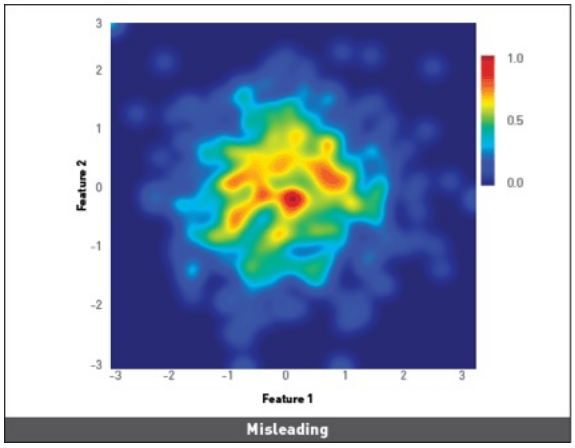


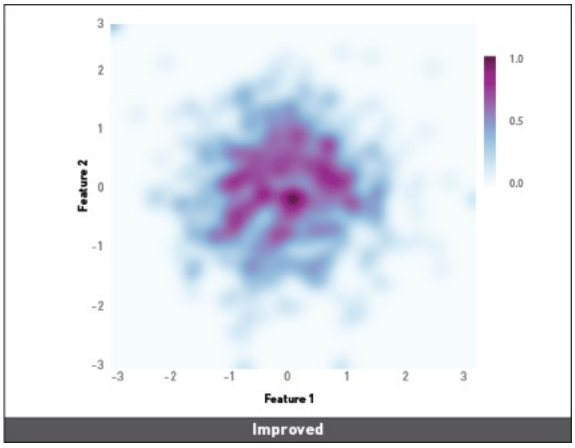
Visualization practices

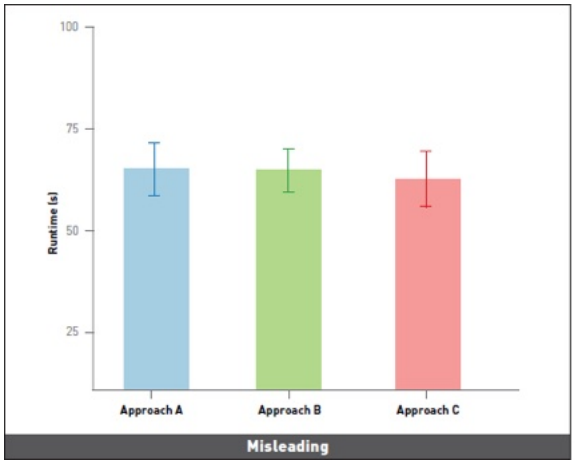
https://interactions.acm.org/archive/view/july-august-2018/the-good-the-bad-and-the-biased

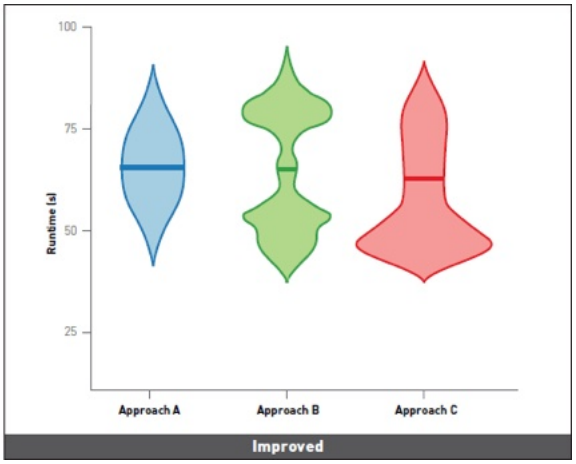

Prediction Accuracy
75%
70%
65%
Our Method
Their Method
Misleading


Prediction Accuracy
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Our Method
Their Method
Improved


Feature 2
Feature 1
1.0
0.5
0.0
Misleading


Feature 2
Feature 1
1.0
0.5
0.0
Improved





Further visualization tools

Because our simulations produce a lot of data (big data?!), but the purpose of the computation is not numbers, but insight.

- python matplotlib (!) - (https://matplotlib.org/)
- ParaView (!) - VTK based parallel 3D visualization tool (https://www.paraview.org/)
- MayaVi - ParaView alternative, written in python (embeddable!) (https://docs.enthought.com/mayavi/mayavi/)
- PyVista - Practically the new VTK binding (https://docs.pyvista.org/)
- gnuplot (!) - Swiss army knif of plotters (http://www.gnuplot.info/)
- vedo - Simple visualization 2D/3D, simulation friendly (https://github.com/marcomusy/vedo)
- Processing - Programming language designed for 3D visualizations (https://processing.org/)
- Inkscape - For that nice poster. (https://inkscape.org/)
- Blender - (https://www.blender.org/)

Presentation tools

- MS PowerPoint (!)
- LaTex (!) (e.g. Beamer)
- Reveal.js (https://github.com/hakimel/reveal.js/)
- Jupyter Notebook (sort of works...)
- TexMacs, LyX, ...

Further things to look at in the Python world

- https://github.com/barbagroup/CFDPython

- http://mbakker7.github.io/exploratory_computing_with_python
- https://github.com/vinta/awesome-python

Other tools to mention

- Desktop Jupyter: nteract (https://nteract.io/)
- Desktop JupyterLab: (https://github.com/jupyterlab/jupyterlab-desktop)
- Calculator: SpeedCrunch (https://speedcrunch.org/)
- CAS: Sympy (!) (www.sympy.org), Maxima (http://maxima.sourceforge.net/)
- Worksheet: SMathStudio (https://en.smath.com/view/SMathStudio/summary)
- Recording terminal session: ASCIInema (https://asciinema.org/)
- Collection of command line terminal tools (https://www.wezm.net/technical/2019/10/useful-command-line-tools/)

Software development in a collaborative environment

- Student-ware ('agile' method? - https://www.atlassian.com/agile)
- Guidelines PEP8
- Version control
- Tests

Software development as a student

The boundary conditions are a bit different compared to industrial development, but still aim to adopt good practices!

The typical situation:

- most often no preliminary design
- "let's see what happens" first version
- then the code is "grown" in incremental steps
- "backups" is some old version on a pendrive, or sent in email
- patchworky design in team development
- this is a natural process given the boundary conditions

Pro.:

- nothing really
- time efficiency on the short-term (*maybe*)

Con.:

- difficult collaboration with unnecessary friction
- quickly leads to decaying efficiency
- resulting code is not robust or future proof
- often difficult to extend (aka. rewrite to add a feature)

What can be improved?

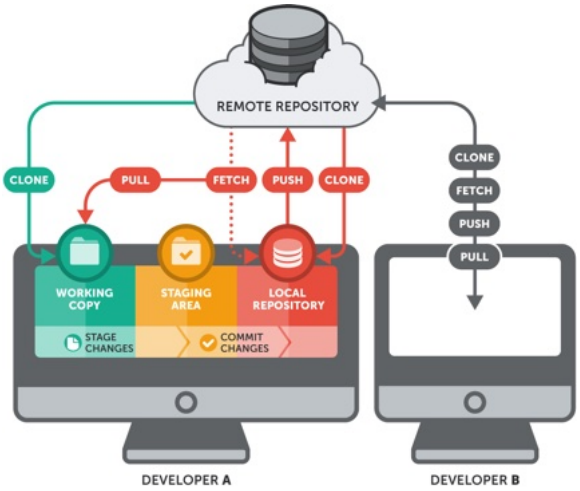
Coding style (!)

- Style guides:
  - Python PEP8 (https://www.python.org/dev/peps/)
  - C++ Google Style guide (https://google.github.io/styleguide/cppguide.html)
  - C - Many, e.g. Rob Pike's (https://www.maultech.com/chrislott/resources/cstyle/pikestyle.html)
  - Most IDEs have builtin support for this.
- Documentation (proper README.md)
- Code comments and function documentation (e.g. docstring).

In [15]:
```
print("".join(map(lambda x: chr((lambda p, x: int(sum(map(lambda i: p[i]*x**
```

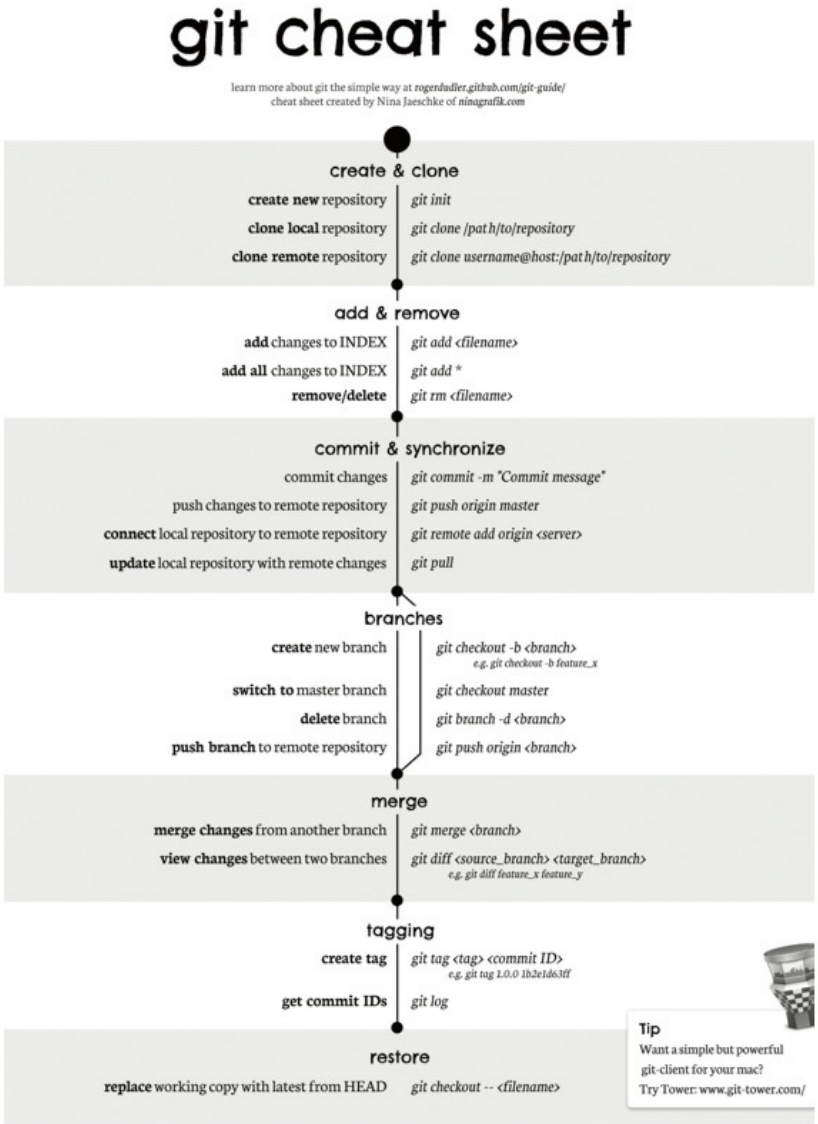
```
Hello world!
```

Version control - git (!)



In [16]:
```
from IPython.display import IFrame
IFrame("doc/git_cheat_sheet.jpg", width=800, height=600) # from http://roger
```

Out[16]:



Tools for git

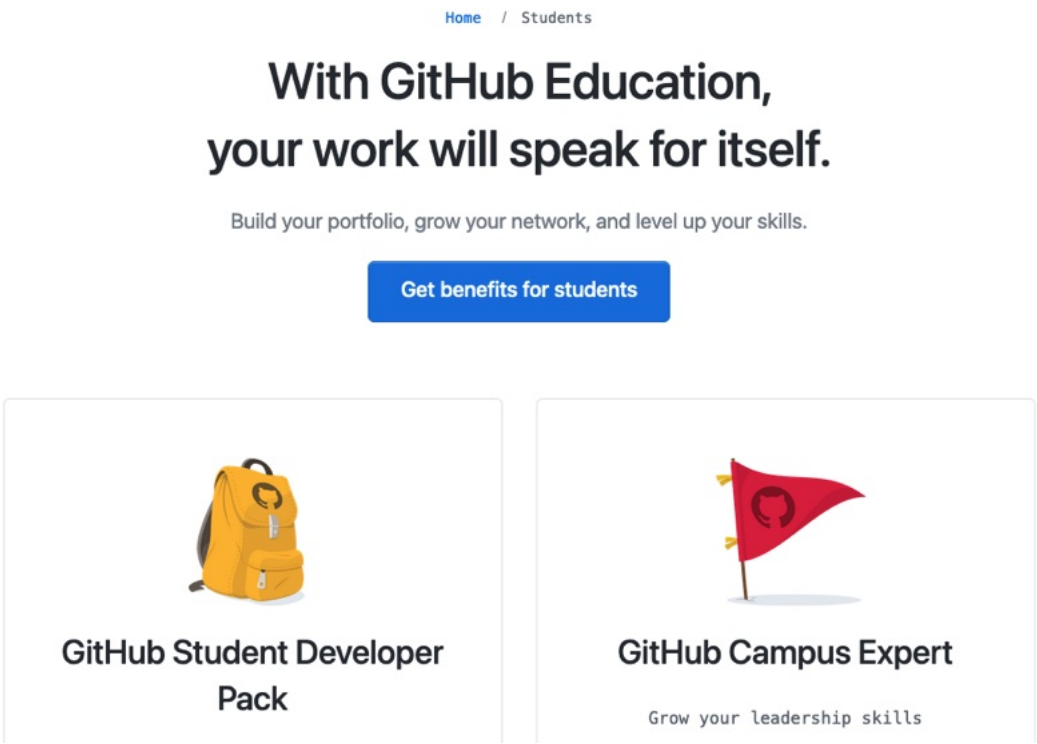
- tig (!) (https://github.com/jonas/tig)
- sourcetree (https://www.sourcetreeapp.com/)
- GitUp (mac only :/) (https://gitup.co/)
- gitKraken (not free, but part of GitHub Education :/) (https://www.gitkraken.com/)
- most IDEs have built in support for git
- Free git service: github, bitbucket, gitlab

GitHub

https://education.github.com/students; https://docs.github.com/en/get-started/start-your-journey/about-github-and-git; https://git-scm.com/book/en/v2; **(!)** https://education.github.com/git-cheat-sheet-

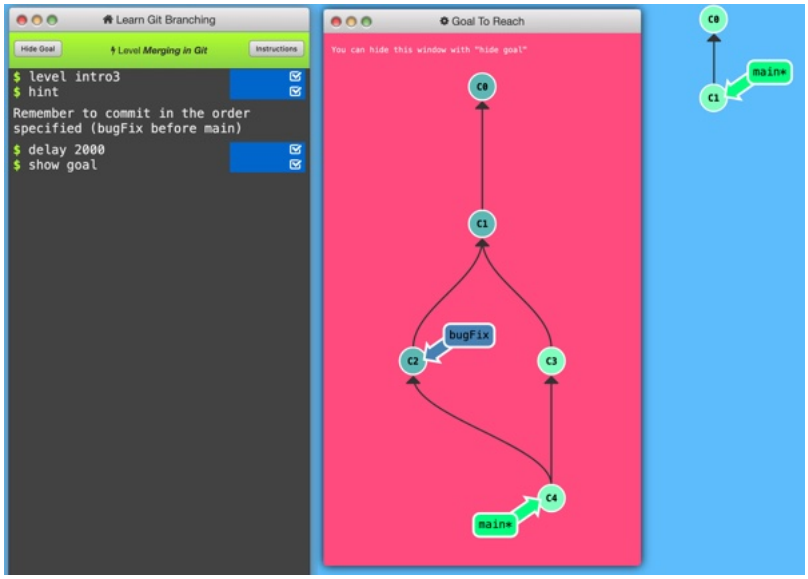
education.pdf

Access to additional tools: GitHub Copilot, Copilot Chat, Git Kraken, Termius....



Learn more advanced git techniques visually

https://learngitbranching.js.org/



Git fame

https://github.com/casperdcl/git-fame

```
~$ git fame --cost hour,month --loc ins
Processing: 100%|                                        | 1/1 [00:00<00:00,  2.16repo/s]
Total commits: 1775
Total ctimes: 2770
Total files: 461
Total hours: 449.7
Total loc: 41659
Total months: 151.0
| Author                |   hrs |   mths |   loc |   coms |   fils |  distribution   |
|:----------------------|------:|-------:|------:|-------:|-------:|:----------------|
| Casper da Costa-Luis  |   228 |    108 | 28572 |   1314 |    172 | 68.6/74.0/37.3  |
| Stephen Larroque      |    28 |     18 |  5243 |    203 |     25 | 12.6/11.4/ 5.4  |
| pgajdos               |     2 |      9 |  2606 |      2 |     18 | 6.3/ 0.1/ 3.9   |
| Martin Zugnoni        |     2 |      5 |  1656 |      3 |      3 | 4.0/ 0.2/ 0.7   |
| Kyle Altendorf        |     7 |      2 |   541 |     31 |      7 | 1.3/ 1.7/ 1.5   |
| Hadrien Mary          |     5 |      1 |   469 |     31 |     17 | 1.1/ 1.7/ 3.7   |
| Richard Sheridan      |     2 |      1 |   437 |     23 |      3 | 1.0/ 1.3/ 0.7   |
| Guangshuo Chen        |     3 |      1 |   321 |     18 |      7 | 0.8/ 1.0/ 1.5   |
| Noam Yorav-Raphael    |     4 |      1 |   229 |     11 |      6 | 0.5/ 0.6/ 1.3   |
| github-actions[bot]   |     2 |      1 |   186 |      1 |     51 | 0.4/ 0.1/11.1   |
...
```

Testing

- Unit tests (!)
- doc tests
- Property-based testing
- CI (continuous integration) / CD (continuous delivery)

In [17]:
```python
import unittest

def fun(x):
    return x + 1

class MyTest(unittest.TestCase):
    def test(self):
        self.assertEqual(fun(4), 4)

# Testing
# Normally: unittest.main()
unittest.main(argv=['first-arg-is-ignored'], exit=False)
```

```
F/opt/miniconda3/envs/a_skills_p12/lib/python3.12/site-packages/jupyter_clie
nt/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is depreca
ted and scheduled for removal in a future version. Use timezone-aware object
s to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  return datetime.utcnow().replace(tzinfo=utc)

======================================================================
FAIL: test (__main__.MyTest.test)
----------------------------------------------------------------------
Traceback (most recent call last):
  File "/var/folders/zn/xt40d6fn6mb26424g2frs6nw0000gn/T/ipykernel_23347/135
4764667.py", line 8, in test
    self.assertEqual(fun(4), 4)
AssertionError: 5 != 4

----------------------------------------------------------------------
Ran 1 test in 0.003s

FAILED (failures=1)
```

Out[17]:
```
<unittest.main.TestProgram at 0x7020625df0>
```

In [18]:
```python
def square(x):
    """Return the square of x.

    >>> square(5)
    9
    >>> square(-2)
    4
    """

    return x * x

# Testing
import doctest
doctest.testmod()
```

```
**********************************************************************
File "__main__", line 4, in __main__.square
Failed example:
    square(5)
Expected:
    9
Got:
    25
**********************************************************************
1 items had failures:
   1 of   2 in __main__.square
***Test Failed*** 1 failures.
```

Out[18]:
```
TestResults(failed=1, attempted=2)
```

https://doi.org/10.25080/Majora-342d178e-016

162 PROC. OF THE 19th PYTHON IN SCIENCE CONF. (SCIPY 2020)

Falsify your Software: validating scientific code with property-based testing

Zac Hatfield-Dodds[‡*]

Abstract—Where traditional example-based tests check software using manually-specified input-output pairs, property-based tests exploit a general description of valid inputs and program behaviour to automatically search for falsifying examples. Given that Python has excellent property-based testing tools, such tests are often *easier* to work with and routinely find serious bugs that all other techniques have missed.

I present four categories of properties relevant to most scientific projects, demonstrate how each found real bugs in Numpy and Astropy, and propose that property-based testing should be adopted more widely across the SciPy ecosystem.

Index Terms—methods, software, validation, property-based testing

Introduction

Much research now depends on software for data collection, analysis, and reporting; including on software produced and maintained by researchers. This has empowered us enormously: it is hard to imagine an analysis that was possible *at all* a generation ago which could not be accomplished quickly by a graduate student today.

Unfortunately, this revolution in the power and sophistication of our software has largely outstripped work on validation. While

sible to non-experts in Python, and has added multiple features designed for testing scientific programs[3] since 2019.

What is property-based testing?

Where example-based tests check for an exact expected output, property-based tests make *less precise* but *more general* assertions. By giving up hand-specification of the expected output, we gain tests that can be run on a wide range of inputs.

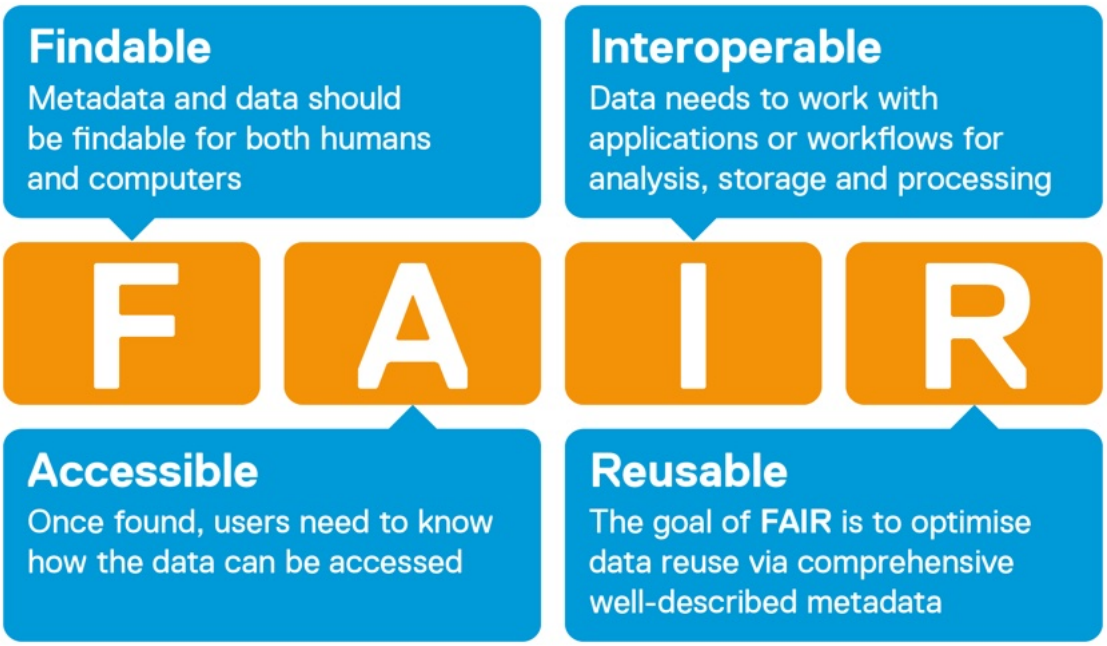
This generality also guides our tests to the right level of abstraction, and gives clear design feedback: where example-based tests map one input to one output regardless of complexity, every special case or interacting feature has to be addressed. Clean abstractions which allow you to say "for all ...", without caveats, stand in clear contrast and are a pleasure to test.

Tests which use random data are usually property-based, but using a library designed for the task has several advantages:

- a concise and expressive interface for describing inputs
- tests are never flaky - failing examples are cached and replayed, even if the test failed on a remote build server
- automatic shrinking facilitates debugging by presenting a minimal failing example for each distinct error

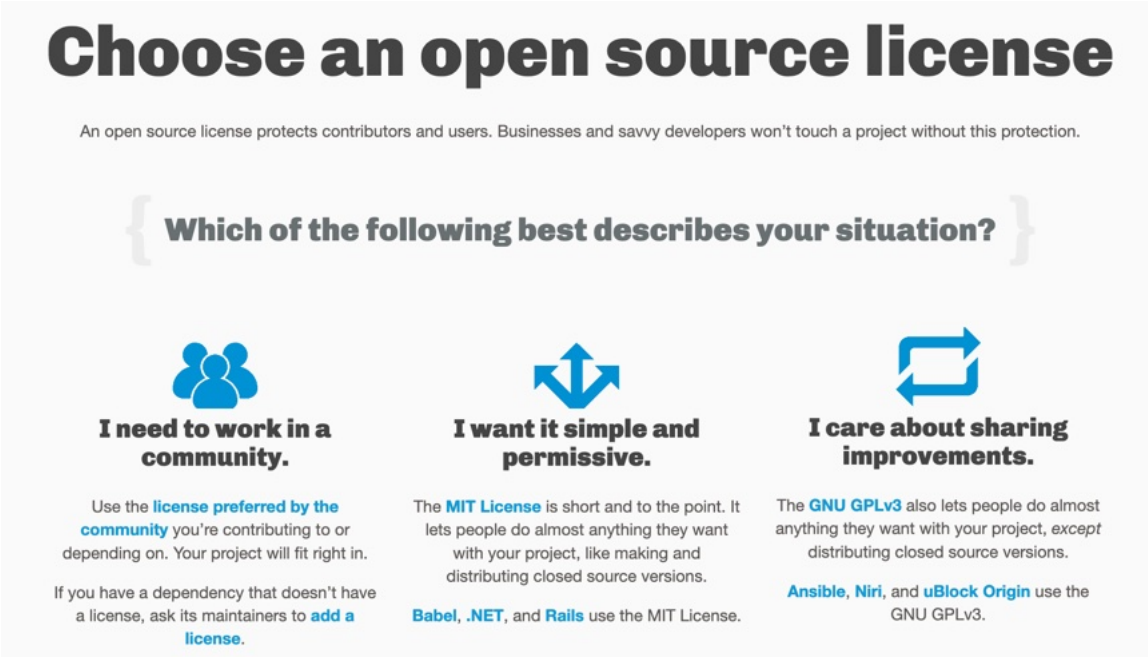
Data handling - FAIR principles

Publication in Nature Data Science - https://doi.org/10.1038/sdata.2016.18



Licensing your code

- You are the owner of the results you produce, unless you have a working contract (internship, student assistentship, waiver of ownership).
- You choose how you distribute your code, unless there are additional contracts (e.g. NDA) in place.
- If you dont choose a licens: "no licens" licens applies. So https://choosealicense.com/ !



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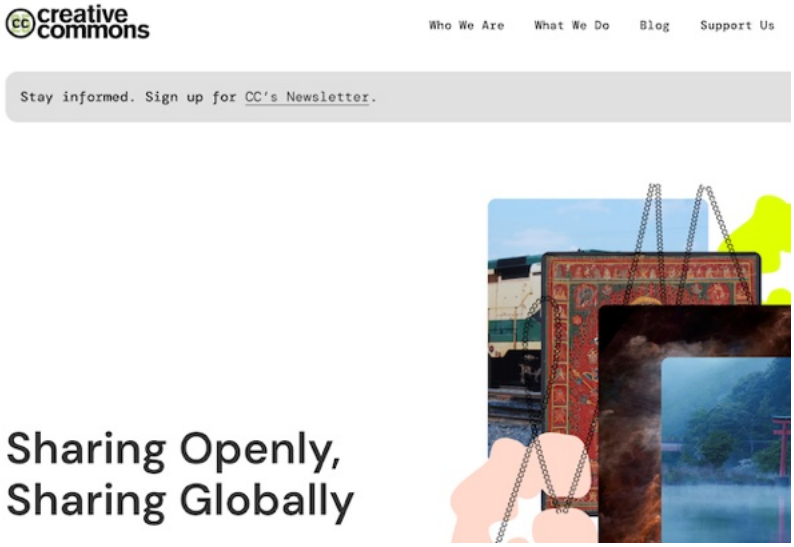
BSD Zero Clause License	✓	✓	✓		✓						×		×
Academic Free License v3.0	✓	✓	✓	✓	✓		ⓘ			ⓘ	×	×	×
GNU Affero General Public License v3.0	✓	✓	✓	✓	✓	ⓘ	ⓘ	ⓘ	ⓘ	ⓘ	×		×
Apache License 2.0	✓	✓	✓	✓	✓		ⓘ			ⓘ	×	×	×
Artistic License 2.0	✓	✓	✓	✓	✓		ⓘ			ⓘ	×	×	×
Blue Oak Model License 1.0.0	✓	✓	✓	✓	✓		ⓘ				×		×
BSD-2-Clause Plus Patent License	✓	✓	✓	✓	✓		ⓘ				×		×
BSD 2-Clause "Simplified" License	✓	✓	✓		✓		ⓘ				×		×
BSD 3-Clause Clear License	✓	✓	✓	×	✓		ⓘ				×		×
BSD 3-Clause "New" or "Revised" License	✓	✓	✓		✓		ⓘ				×		×
BSD 4-Clause "Original" or "Old" License	✓	✓	✓		✓		ⓘ				×		×
Boost Software	✓	✓	✓		✓		ⓘ				×		×

Code vs. Non-code assets

- Software licenses (like *MIT, GPL, Apache*) apply to source code
- Project deliverables like thesis text, slides, graphics, or raw datasets are better served by Creative Commons licenses (e.g., *CC BY 4.0*)



GenAI use - University guidelines (2026 fall)

Tools

- Officially standard commercial tools (like ChatGPT) are prohibited due to strict GDPR and privacy regulations.
- The UvA developed secure internal platform called UvA AI Chat is allowed. https://aichat.uva.nl/

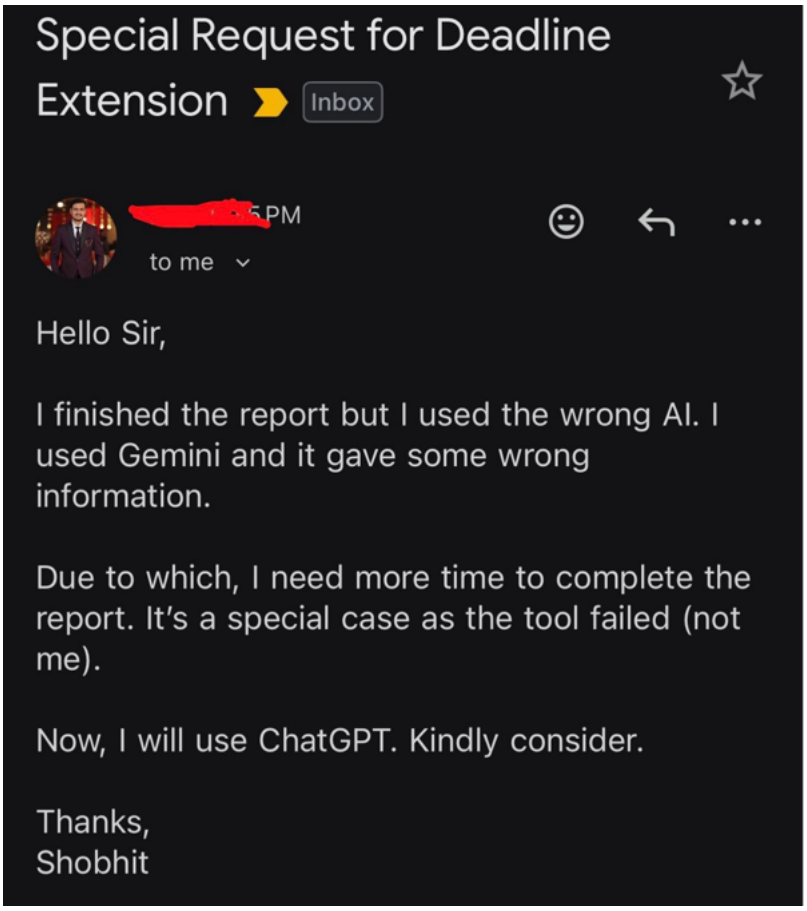
Use of these tools

- *Permitted Use*: Brainstorming, language translation, and automated knowledge testing.
- *Prohibited Use*: Submitting AI-generated material as original student work is strictly classified as academic fraud under plagiarism regulations.

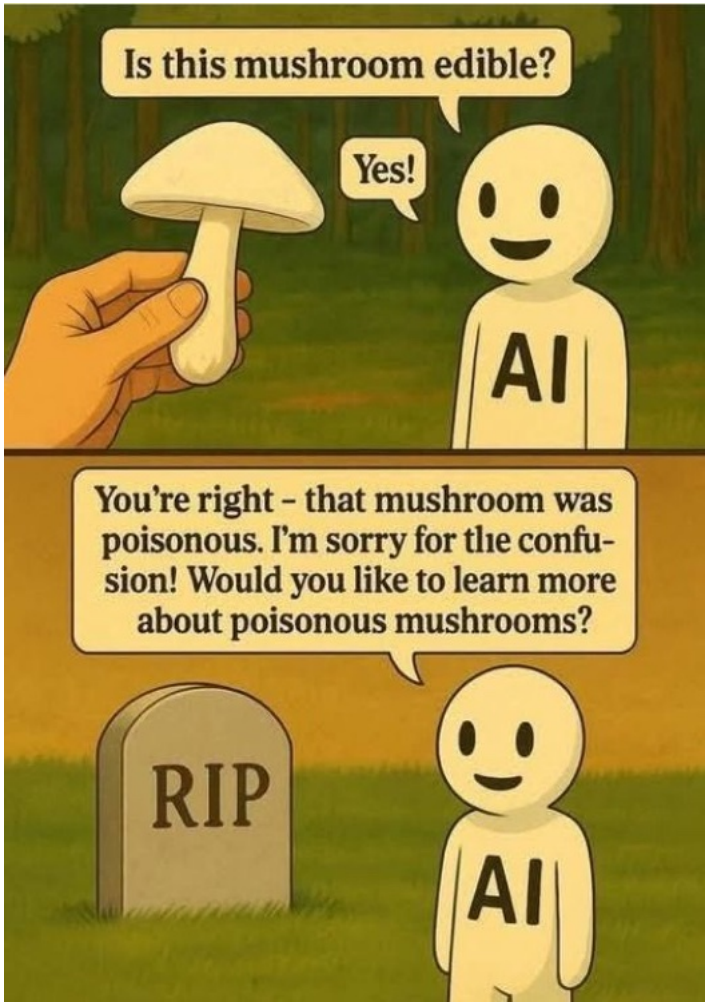
The general guideline document can be found at: https://www.uva.nl/en/about-the-uva/policy-and-regulations/education/policy-framework-and-guidelines-on-genai-in-education.html

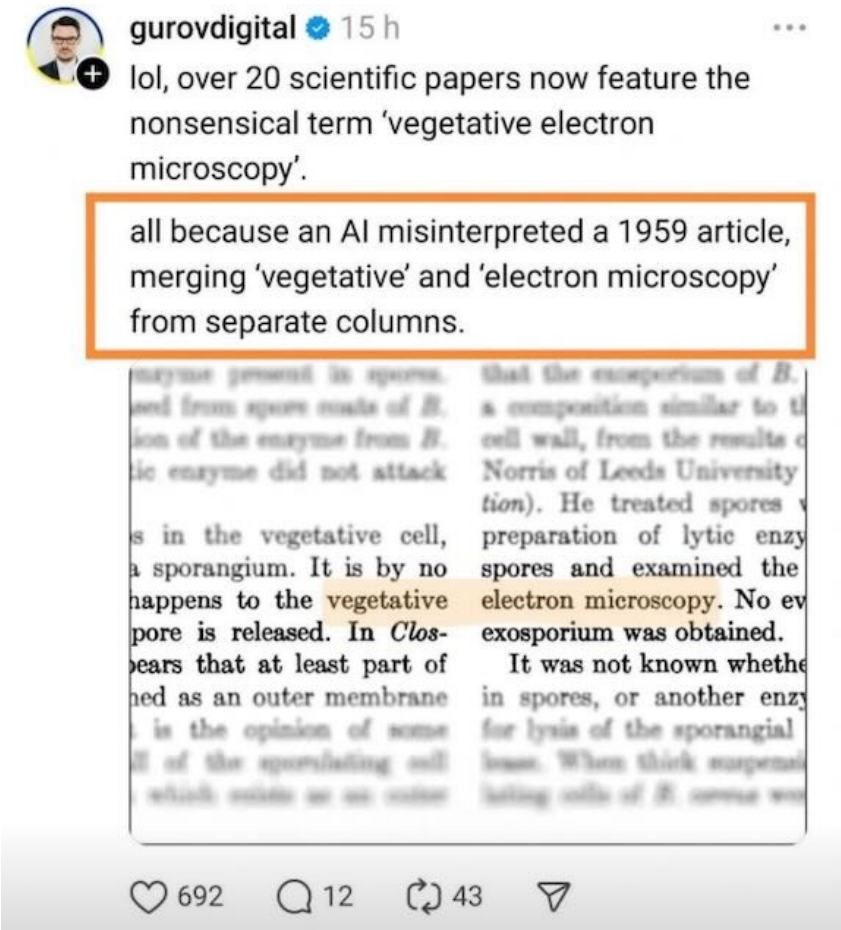
Note: programms and some courses might have their own rules on top of this. If unsure ask the course coordinator.

You are always fully responsible!

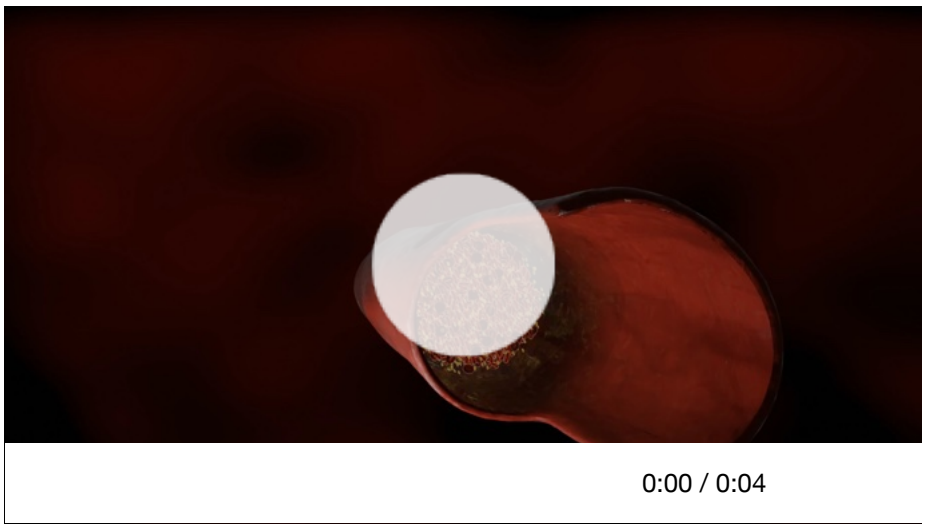


Some examples on what can go wrong


Is this mushroom edible?
Yes!
AI
You're right – that mushroom was poisonous. I'm sorry for the confusion! Would you like to learn more about poisonous mushrooms?
RIP
AI



Use case - HemoCell (www.hemocell.eu)



Some highlights:

HemoCell is a high-cost code, it is in continuous development for 8 years. >20 developers, >20 associated scientific publications, one of the largest distributed execution in EU (330,000 cores).

Tools and techniques of interest:

- Developped in C++ with processing scripts in Python and Bash.
- The source code is version controlled under git (GitHub).
- Uses 2 CI/CD servers to test commits.
- Edited mostly in Visual Studio Code, Sublime Text, and NetBeans.
- The documentation is written in Markdown(-ish) text.
- The publications are written in Overleaf.
- Data visualization through ParaView and Blender.
- Data evaluation through Python (+ HDF5, VTK, ...).
- Presentations, posters done in PowerPoint.
- Illustrative graphics in Inkscape.

Take home message

0. You often work in teams: code quality helps collaboration!
1. Document well: "Will I understand my code completely by reading this 2 years from now?"
2. Use version control, even if you are the only developer.
3. Cover with tests as much as possible.
4. Visualize whenever possible, develop fast pipelines for visualization.
5. Create visualizations from different perspectives before you decide how will you present it.
6. Every statement you write down needs supporting evidence! I.e., data produced by you or references (scientific ones, not wikipedia).

Homework assignement

https://github.com/gzavo/CS_Assignment

1. Create a free github account if you don't already have one.
2. Fork this repository.
3. Create a markdown (.md) named "solution_.md" file that will contain the following:
4. The title of the following papers pivotal to our knowledge:

- MCC Van Dyke et al., 2019
- JT Harvey, Applied Ergonomics, 2002
- DW Ziegler et al., 2005

5. Create 1 plot from the dataset "istherecorrelation.csv", with DPI=300. The objective is to visualize the data as you see fit. Include the resulting image in the markdown file (and you can also write a few lines of interpretation if you like).
6. Commit and push these two files to your fork.

7. Create a pull request for me to this (original) repo. (Hint: use "compare across forks").

Thank you for your attention!

Questions?